

Problem 1

```
rgb = imread('lena.bmp');  
figure;  
imshow(rgb);  
title('Original lena.bmp')
```

Original lena.bmp



```
gray = rgb2gray(rgb);  
figure;  
imshow(gray)  
title('Grayscale lena.bmp (Built-In Function)')
```

Grayscale lena.bmp (Built-In Function)



```
my_gray = grayscale_intensity(rgb);  
figure;  
imshow(my_gray)  
title('Grayscale lena.bmp (User Function)')
```

Grayscale lena.bmp (User Function)



```
figure;  
imwrite(my_gray, 'lena_gray.jpg')
```

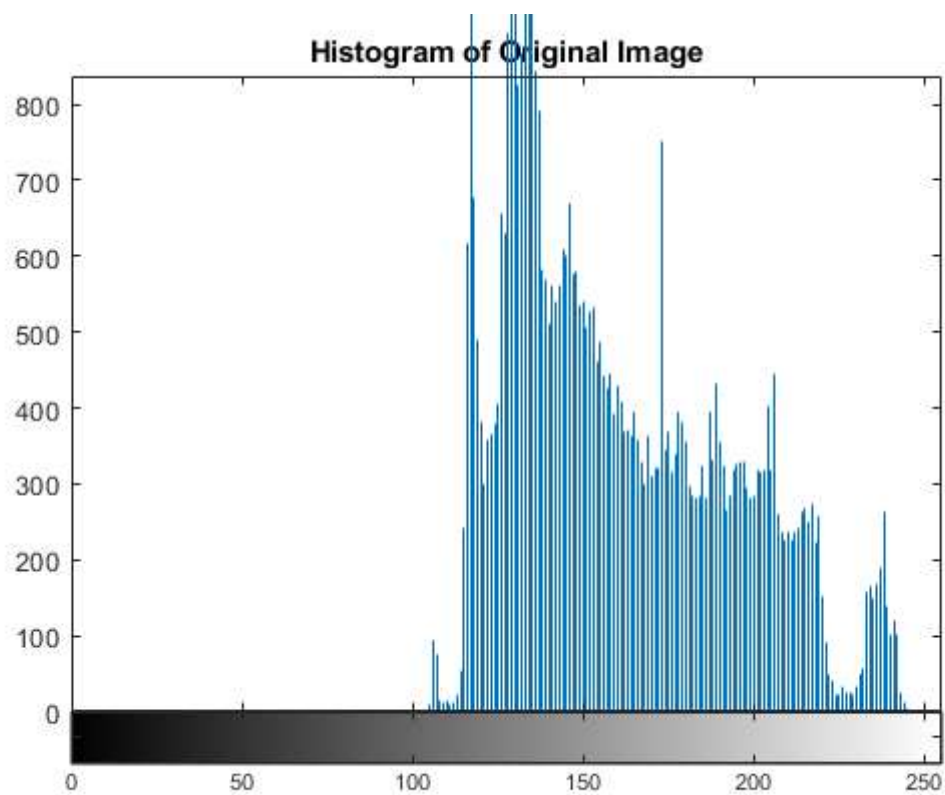
Problem 2

```
img = imread('lowcontrast.jpg');  
figure;  
imshow(img)  
title('Original Low-Contrast Image')
```

Original Low-Contrast Image



```
figure;  
imhist(img)  
title('Histogram of Original Image')
```

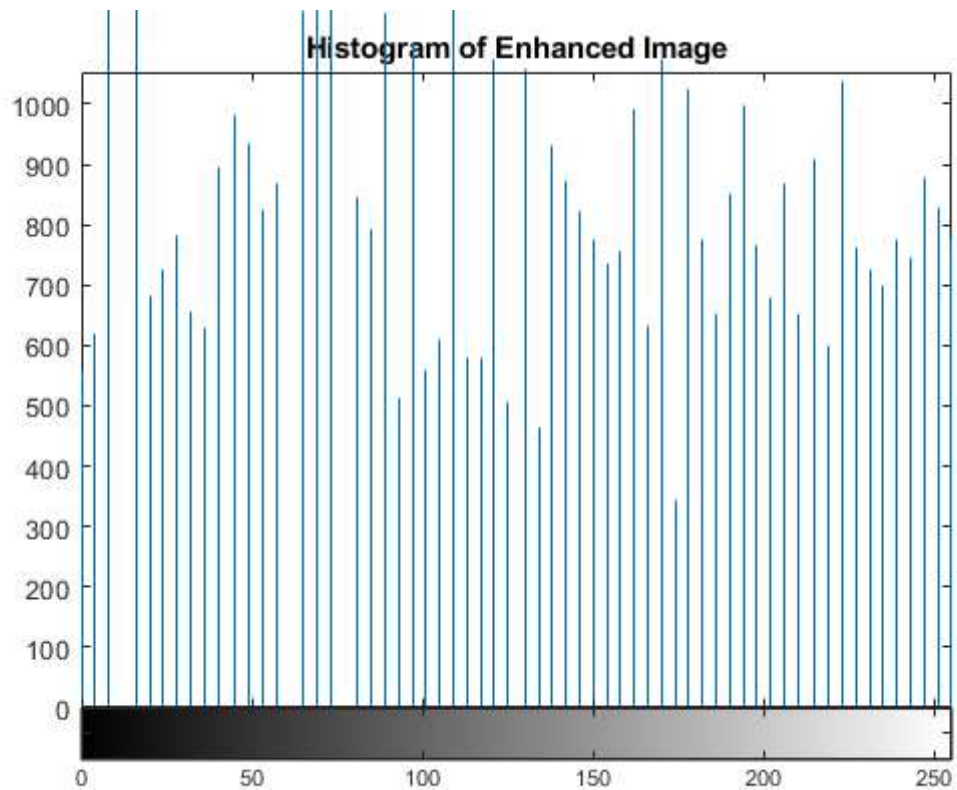


```
newcmap = histeq(img);  
  
figure;  
imshow(newcmap)  
title('Enhanced Low-Contrast Image')
```

Enhanced Low-Contrast Image



```
figure;  
imhist(newcmap)  
title('Histogram of Enhanced Image')
```



Problem 3

```
noisy_lena = imnoise(gray, 'salt & pepper', 0.05);  
  
figure;  
imshow(noisy_lena)
```

```
title('lena.bmp with Salt and Pepper Noise')
```

lena.bmp with Salt and Pepper Noise



```
filtered_3x3 = medfilt2(noisy_lena, [3 3]);
```

```
figure;  
imshow(filtered_3x3)  
title('Filtered Image (Median 3x3)')
```

Filtered Image (Median 3x3)



```
filtered_5x5 = medfilt2(noisy_lena, [5 5]);  
  
figure;  
imshow(filtered_5x5)  
title('Filtered Image (Median 5x5)')
```

Filtered Image (Median 5x5)



```
% The 5x5 filtered image seems to be more blurry than that of the 3x3  
% filtered image. As the window size of the median filter increases,  
% the image gets increasingly blurry.
```

Functions

```
function my_gray = grayscale_intensity(rgb)  
    R = rgb(:,:,1);  
    G = rgb(:,:,2);  
    B = rgb(:,:,3);  
  
    my_gray = (0.3*R) + (0.6*G) + (0.1*B);  
  
    my_gray = uint8(my_gray);  
end
```

